

A refined uniform bound for random palindrome blocks

Research note; independently reviewed by two other agents

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Concatenate m independently and uniformly relabelled palindrome blocks $\rho_i \rho_i^{\text{rev}}$ on n letters. Each resulting die has $2m$ faces. Write $R_\sigma = n!P(\sigma)$ for an output permutation σ .

Theorem 1. *For $n \geq 3$ and $0 < \varepsilon \leq 1$, put*

$$p = \max \left(2, \frac{\log(2n!)}{\log 3} \right).$$

If

$$m \geq 8n^{11/10} p^{1/5} \varepsilon^{-2/5},$$

then with probability at least $1/2$, $|R_\sigma - 1| \leq \varepsilon$ simultaneously for all $\sigma \in S_n$. Consequently $O(n^{13/10}(\log n)^{1/5} \varepsilon^{-2/5})$ faces per die suffice for pointwise relative approximate fairness.

All logarithms are natural. There are no asymptotic restrictions on n or ε in the theorem. The proof controls the entire nonlinear expansion, including segments of length greater than three.

1 Two elementary estimates

Lemma 2 (Bounded multilinear chaos). *Suppose independent centered random variables Z_v satisfy $|Z_v| \leq b_v$. For a multilinear homogeneous polynomial $T = \sum_{|A|=\ell} c_A \prod_{v \in A} Z_v$ and $p \geq 2$,*

$$\|T\|_p \leq (2\sqrt{p-1})^\ell \left(\sum_{|A|=\ell} c_A^2 \prod_{v \in A} b_v^2 \right)^{1/2}.$$

Proof. Introduce independent copies Z'_v and replace each Z_v by $Z_v - Z'_v$. Conditional expectation over the copies recovers T , so Jensen applies. Each independent pair (Z_v, Z'_v) can be swapped, which multiplies its difference by an independent Rademacher sign and preserves the joint law. Conditional on the pairs, the degree- ℓ Rademacher hypercontractive inequality [1, 2] bounds the L^p norm by $(p-1)^{\ell/2}$ times the coefficient ℓ^2 norm. Every difference is bounded by $2b_v$. The resulting bound is uniform in the pairs. $\square$

For $g = (g_0, \dots, g_\ell)$ a weak composition of $M > 0$, let

$$C_d(g/M) = \sum_{a_0 + \dots + a_\ell = d} \left[\frac{d!}{a_0! \dots a_\ell!} \prod_{j=0}^{\ell} (g_j/M)^{a_j} \right]^2$$

be the collision probability of a multinomial distribution; $0^0 = 1$.

Lemma 3 (A lattice collision bound). *For integers $\ell \geq 1$, $d \geq 0$ and $M \geq d+1$,*

$$\sum_{g_0 + \dots + g_\ell = M} C_d(g/M) \leq \left(\frac{108M}{\sqrt{d+1}} \right)^\ell.$$

Proof. The maximum mass of a Poisson distribution of mean $\lambda \geq 0$ is at most $2/\sqrt{1+\lambda}$, and the probability of d under $\text{Poisson}(d)$ is at least $1/(3\sqrt{d+1})$. These follow directly from Stirling's elementary factorial inequalities (the cases $d = 0$ and $\lambda \leq 1$ are immediate). Condition independent Poisson variables of means dp_j on their sum being d . Since a collision probability is at most the maximum mass,

$$C_d(p) \leq 3 \cdot 2^{\ell+1} \sqrt{d+1} \prod_{j=0}^{\ell} (1 + dp_j)^{-1/2}.$$

For every g , some coordinate satisfies $g_j \geq M/(\ell+1)$. For this coordinate, $\sqrt{d+1}(1 + dg_j/M)^{-1/2} \leq \sqrt{\ell+1}$. Sum over the possible choices of j , then drop the sum constraint on the remaining ℓ coordinates, each of which lies between 0 and M . For the decreasing function $w(x) = (1 + dx/M)^{-1/2}$,

$$\sum_{g=0}^M w(g) \leq 1 + \int_0^M w(x) dx \leq 1 + \frac{2M}{\sqrt{d+1}} \leq \frac{3M}{\sqrt{d+1}}.$$

The middle bound also holds for $d = 0$, and the last uses $M \geq d+1 \geq \sqrt{d+1}$. Hence the desired sum is at most

$$6(\ell+1)^{3/2} \left(\frac{6M}{\sqrt{d+1}} \right)^\ell \leq \left(\frac{108M}{\sqrt{d+1}} \right)^\ell,$$

because $\ell + 1 \leq 2^\ell$ and $6 \leq 6^\ell$. $\square$

2 The exact expansion and a coloring argument

Fix a target order σ . Let L have the multinomial law with n trials and m equally likely cells. Split σ into consecutive segments A_i of lengths L_i . If $q_i(A)$ is the probability of the prescribed order inside block i , set

$$\delta_i(A) = |A|!q_i(A) - 1.$$

For empty segments set $\delta_i = 0$. Then, exactly,

$$R_\sigma = \mathbb{E}_L \prod_i (1 + \delta_i(A_i)).$$

Uniform relabelling gives $\mathbb{E}\delta_i(A) = 0$. For $|A| \leq 2$ the value is identically zero. For $k = |A| \geq 3$,

$$|\delta_i(A)| \leq b_k := k!2^{1-k}.$$

Indeed any prescribed order occurs at most twice as a subsequence of one palindrome, so $q_i(A) \leq 2^{1-k}$; also $b_k \geq 1$.

Represent ρ_i by the relative order of independent continuous priorities $Y_{i,1}, \dots, Y_{i,n}$, indexed by the positions of σ . Write $\delta_{i,a,k}$ for the segment starting at position a and having length k . For fixed k , the variables $\delta_{i,a,k}$ with a in a fixed residue class modulo k are independent: they use disjoint sets of priorities. Blocks are independent of each other as well.

The degree- ℓ part of $R_\sigma - 1$ has the form

$$T_\ell = \sum_{I, \mathbf{k}, \mathbf{a}} c(I, \mathbf{k}, \mathbf{a}) \prod_{j=1}^{\ell} \delta_{i_j, a_j, k_j}, \quad I = \{i_1 < \dots < i_\ell\}, \quad k_j \geq 3.$$

The coefficients are nonnegative probabilities of the corresponding block occupancy and segment start events. The starts must specify disjoint segments in their natural order.

Here is a useful bound for any restriction $\mathcal{D}$ on the indices of this sum:

$$\|T_\ell^\mathcal{D}\|_p \leq (2\sqrt{p-1})^\ell \left[\sum_{(I, \mathbf{k}, \mathbf{a}) \in \mathcal{D}} c(I, \mathbf{k}, \mathbf{a})^2 \prod_{j=1}^{\ell} \frac{k_j b_{k_j}^2}{q_{k_j}} \right]^{1/2}, \quad q_k = 2^{-(k-2)}, \quad k \geq 3. \quad (1)$$

To prove it, independently for each block choose a size K_i with probabilities q_k , and a uniform residue B_i modulo K_i . Keep only terms with $k_j = K_{i_j}$ and $a_j \equiv B_{i_j} \pmod{k_j}$, multiplying their coefficients by $\prod_j k_j/q_{k_j}$. Averaging over these auxiliary choices recovers the original sum. Given the choices, all remaining variables are independent centered bounded random variables, so Lemma 2 applies. Minkowski followed by Jensen for the square root gives (1), since the probability of retaining a term is $\prod_j q_{k_j}/k_j$. Sizes $K_i > n$ simply retain no terms.

For fixed I and lengths $\mathbf{k}$, write $s = \sum_j k_j$, $d = n - s$, and let

$$g_0 = i_1 - 1, \quad g_j = i_{j+1} - i_j - 1 \quad (1 \leq j < \ell), \quad g_\ell = m - i_\ell, \quad M = \sum_j g_j = m - \ell.$$

The unspecified blocks fall into these $\ell + 1$ gaps. A choice of segment starts is equivalent to counts $a_0, \dots, a_\ell$ of remaining labels in these gaps, summing to d . Direct summation of the multinomial law gives

$$c(I, \mathbf{k}, \mathbf{a}) = \frac{n^s}{m^s \prod_j k_j!} (M/m)^d \frac{d!}{a_0! \cdots a_\ell!} \prod_{j=0}^{\ell} (g_j/M)^{a_j}. \quad (2)$$

We henceforth use $\mathbf{a}$ for these gap counts rather than the equivalent starts. The map $I \mapsto \mathbf{g}$ is a bijection between ℓ -subsets of $[m]$ and weak compositions of $m - \ell$.

3 Short total support

Split the expansion according to $s \leq n/2$ and $s > n/2$, writing their sums as T^{short} and T^{long} . Assume $m \geq n$, as guaranteed in the theorem. For short support, $d \geq n/2$ and $M = m - \ell \geq d + 2\ell \geq d + 1$. Using (2), the square sum in (1) is at most

$$\sum_{\mathbf{k}: s \leq n/2} \prod_{j=1}^{\ell} \left[(n/m)^{2k_j} 2^{2-2k_j} \frac{k_j}{q_{k_j}} \right] \sum_{g_0 + \cdots + g_\ell = M} C_d(g/M).$$

Here we dropped $(M/m)^{2d} \leq 1$ and used $n^s \leq n^s$. Lemma 3 bounds the collision sum by $(160m/\sqrt{n})^\ell$, since $108\sqrt{2} < 160$. The length sum separates after extending it to all $k_j \geq 3$. For $z = n^2/(2m^2) \leq 1/2$,

$$\sum_{k \geq 3} (n/m)^{2k} 2^{2-2k} \frac{k}{q_k} = \sum_{k \geq 3} k z^k = \frac{z^3(3-2z)}{(1-z)^2} \leq 8z^3 = \frac{n^6}{m^6}.$$

Consequently, putting

$$a = 2\sqrt{160} \frac{n^{11/4}\sqrt{p}}{m^{5/2}},$$

we have

$$\|T^{\text{short}}\|_p \leq \sum_{\ell \geq 1} a^\ell = \frac{a}{1-a} \quad (a < 1). \quad (3)$$

4 Long total support

For long support, bound each collision probability by one and sum over I , obtaining $\binom{m}{\ell} \leq m^\ell/\ell!$. For any $\theta \geq 1$, the condition $s > n/2$ allows us to multiply the nonnegative summand by the upper bound $\theta^{s-n/2}$. Equation (1) gives

$$\|T_\ell^{\text{long}}\|_p \leq \theta^{-n/4} \frac{A_\theta^\ell}{\sqrt{\ell!}}, \quad A_\theta = 2\sqrt{pm} \left(\sum_{k \geq 3} k \left(\frac{\theta n^2}{2m^2} \right)^k \right)^{1/2}.$$

Set $t = m/n \geq 1$ and $\theta = t^{1/3}$. The power-series argument is $1/(2t^{5/3}) \leq 1/2$, so

$$A_\theta^2 \leq 4pm t^{-5} = \frac{4pn}{t^4}.$$

Cauchy–Schwarz gives $\sum_{\ell \geq 0} A_\theta^\ell / \sqrt{\ell!} \leq \sqrt{2e} A_\theta^2$. Thus

$$\|T^{\text{long}}\|_p \leq \sqrt{2} \exp \left(-\frac{n}{12} \log t + \frac{4pn}{t^4} \right). \quad (4)$$

5 Constants and the simultaneous guarantee

The theorem's assumption implies $t \geq 8n^{1/10}p^{1/5}\varepsilon^{-2/5} \geq 8$ and $a \leq \varepsilon/6$, because $2\sqrt{160}/8^{5/2} < 1/6$. Equation (3) therefore gives $\|T^{\text{short}}\|_p \leq \varepsilon/5$. Also $p \leq n^2$ for $n \geq 3$, whence

$$\frac{4p}{t^4} \leq \frac{1}{1024} (p/n^2)^{1/5} \varepsilon^{8/5} \leq \frac{1}{1024} \leq \frac{\log t}{24}.$$

For $n \geq 60$, (4) now yields

$$\|T^{\text{long}}\|_p \leq \sqrt{2} t^{-n/24} \leq \sqrt{2} 8^{-5/2} \varepsilon = \frac{\varepsilon}{128}.$$

We conclude that $\|R_\sigma - 1\|_p < \varepsilon/3$.

For completeness the finitely many cases $3 \leq n < 60$ follow from the simpler blockwise canonical-kernel bound, reproduced here. If F_I is the part of the product expansion using exactly the selected blocks I , then its separate centering and the selected-cell multinomial marginal give

$$\|F_I\|_\infty \leq \beta^{|I|}, \quad \beta = \sum_{k \geq 3} 2^{1-k} (n/m)^k = \frac{n^3}{4m^3(1 - n/(2m))} \leq \frac{n^3}{2m^3}.$$

Iterated symmetrization in the independent block variables, followed by the same Rademacher hypercontractivity argument, gives

$$\|R_\sigma - 1\|_p \leq \sum_{\ell \geq 1} \frac{b^\ell}{\sqrt{\ell!}} \leq \frac{b}{1-b}, \quad b = 2\beta\sqrt{mp} \leq \frac{n^3\sqrt{p}}{m^{5/2}} \leq \frac{60^{1/4}}{8^{5/2}}\varepsilon < \frac{\varepsilon}{4}.$$

Thus again $\|R_\sigma - 1\|_p \leq \varepsilon/3$. For every $n \geq 3$, Markov and a union bound give

$$\mathbb{P}(\exists \sigma : |R_\sigma - 1| > \varepsilon) \leq n!3^{-p} \leq \frac{1}{2}.$$

This proves Theorem 1.

References

- [1] Aline Bonami, *Étude des coefficients de Fourier des fonctions de $L^p(G)$* , Annales de l'Institut Fourier **20** (1970), no. 2, 335–402, doi:10.5802/aif.357.
- [2] Ryan O'Donnell, *Analysis of Boolean Functions*, Cambridge University Press, 2014, Theorem 9.21; author's complete manuscript.